Md Ahsanul Haque, Ph.D.
MAT-120

Quiz-4

Date: April 28, 2026

Name

Section

Instructions:

- Write your full name and section number.
- This quiz contains 2 problems. It is **YOUR responsibility** to check that your copy has all 2 problems.
- Make sure to read the instructions for each problem carefully.
- Use correct mathematical notation.
- Write cohesively and follow a logical sequence of steps.
- Check that your answer can be read and understood by the grader.

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1. (10 points) The radius of a right circular cylinder is decreasing at a constant rate of 0.2 cm/s while its height is increasing at 0.4 cm/s.
   (a) Write the rate of change of the volume with respect to time.
   (b) Determine whether the volume is increasing or decreasing when $r = 5$ cm and $h = 10$ cm.

2. (10 points) Evaluate the following indefinite integrals:
   (a) $\displaystyle\int \left(e^{3x} - 2\right)^2 e^{3x}\, dx$
   (b) $\displaystyle\int \frac{1}{x\sqrt{\ln x}}\, dx$